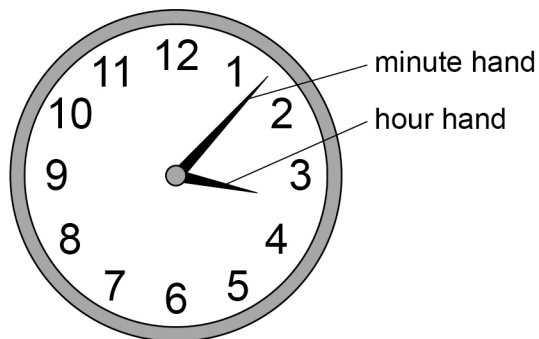


The diagram shows a twelve-hour analogue clock.



What is the average angular speed of the hour hand?

[1 mark]

A $7.3 \times 10^{-5} \text{ rad s}^{-1}$

☐

B $1.5 \times 10^{-4} \text{ rad s}^{-1}$

☐

C $1.8 \times 10^{-3} \text{ rad s}^{-1}$

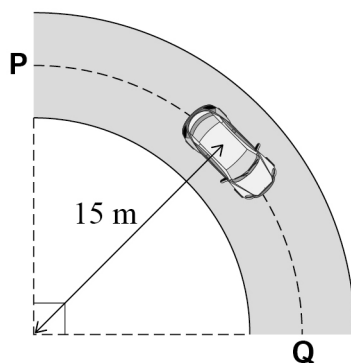
☐

D $5.2 \times 10^{-1} \text{ rad s}^{-1}$

☐

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A car of mass 1200 kg travels at a constant speed around a horizontal circular bend of radius 15 m.



The car travels from **P** to **Q** in 1.8 s.

What is the centripetal force acting on the car?

[1 mark]

A 220 000 N

☐

B 14 000 N

☐

C 5600 N

☐

D 1000 N

☐

Turn over for the next question

Turn over ►



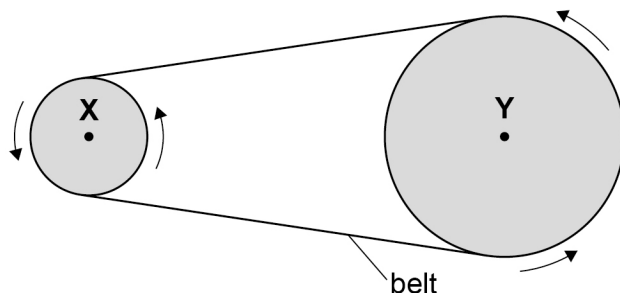
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8

The diagram shows two connected wheels, **X** and **Y**.

The diameter of wheel **Y** is twice the diameter of wheel **X**.
Wheel **X** drives wheel **Y** with a belt. The belt does not slip.
Wheel **X** rotates at a constant rate.



What is $\frac{\text{angular speed of X}}{\text{angular speed of Y}}$?

[1 mark]

A 4

☐

B 2

☐

C 1

☐

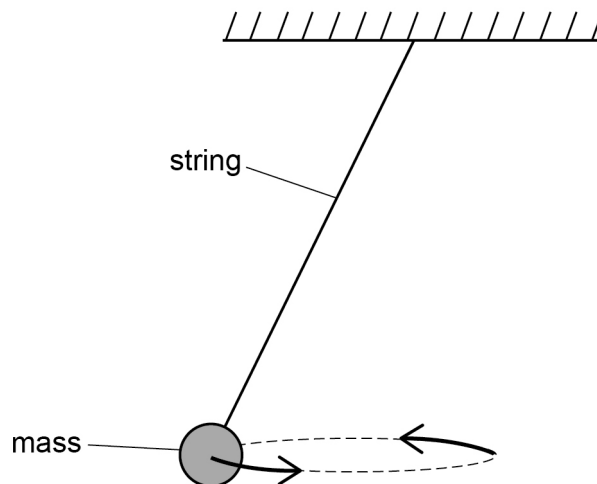
D 0.5

☐

A mass of 0.25 kg is suspended on a string and rotates in a horizontal circle.

The diameter of the circle is 60 cm .

The mass completes one revolution every 1.4 s .



What is the centripetal force acting on the mass?

[1 mark]

A 0.5 N

☐

B 1.5 N

☐

C 2.1 N

☐

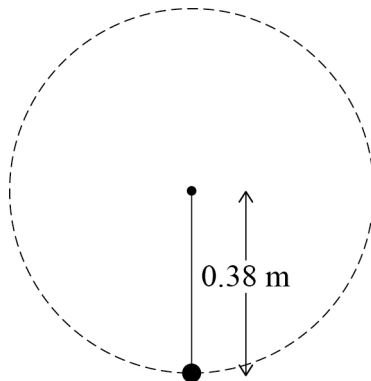
D 3.0 N

☐

Don't waste

2	0
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An object of mass 1.5 kg is attached to the end of a string. The string has a cross-sectional area of 0.79 mm^2 . The object is rotated in a vertical circle at a constant angular speed of 7.5 rad s^{-1} . The radius of the circle is 0.38 m .



What is the stress in the string when the object is at the bottom of the circle?

[1 mark]

A 22 kPa

☐

B 22 MPa

☐

C 59 kPa

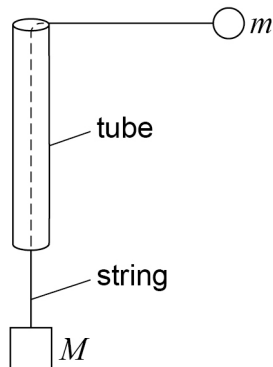
☐

D 59 MPa

☐

In an experiment to investigate centripetal force, mass M and mass m are attached to the ends of a string.

Mass m is moved in a horizontal circle of constant radius.



Mass M is varied as the independent variable.

Which could be a directly measurable dependent variable?

[1 mark]

A speed of m

☐

B centripetal acceleration of m

☐

C mass of m

☐

D time period of rotation of m

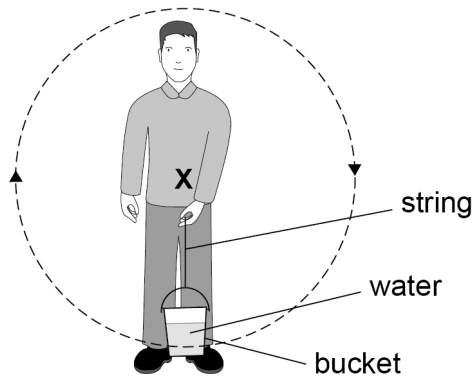
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01

Figure 1 shows a student moving a bucket of water in a vertical circle at a constant speed. The centre of the circle is marked with an **X**. The bucket is attached to a length of string.

Figure 1



01

1

The water moves in a circle whose centre is at **X**.

Draw, on **Figure 1**, an arrow to show the direction of the resultant force acting on the water.

[1 mark]

01

2

At the top of the circle, the angular speed is just sufficient to keep the water in the bucket.

Explain why the centripetal force on the water is equal to its weight at the top of the circle.

[1 mark]



0 1 . 3 The radius of the circle is 0.85 m.

Calculate the angular speed of the water.

[3 marks]

angular speed = _____ rad s⁻¹

0 1 . 4 The student gradually increases the angular speed until the string breaks.

The student plans to repeat the procedure. He wants to achieve a greater angular speed without breaking the string.

He plans to use:

- a new piece of the same type of string
- the same bucket.

Discuss **two** changes the student can make to move water in a vertical circle at a greater angular speed without the string breaking.

[4 marks]

1 _____

2 _____



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ANSWER IN THE SPACES PROVIDED**



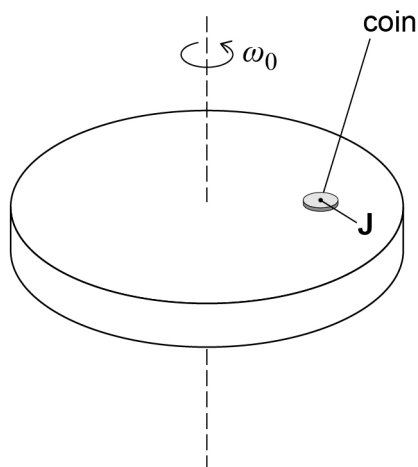
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07

Figure 13 shows a coin on a plate. The centre of mass of the coin is at point **J**.

The plate rotates about its centre in a horizontal plane. The plate initially has a constant angular speed ω_0 . The coin does not slip.

Figure 13



07.1

Draw an arrow on **Figure 13** to show the direction of the frictional force acting on the coin.

[1 mark]

The friction acting on the coin increases with angular speed up to a maximum value. The maximum friction is 85% of the weight of the coin at all points on the plate.

07.2

The angular speed of the plate is gradually increased. Initially, the centre of mass of the coin remains at **J**. At this time it is moving in a circle of circumference 0.94 m.

When the plate reaches an angular speed ω_1 the coin begins to slide.

At ω_1 the friction has reached its maximum value.

Calculate ω_1 .

[3 marks]

$\omega_1 =$ _____ rad s^{-1}



0 7 . 3

The plate is brought to rest and the coin is now placed so that its centre of mass is closer to the centre of the plate.

The angular speed of the plate is gradually increased. The coin begins to slide when the plate reaches an angular speed ω_2 .

Compare ω_2 with ω_1 .

Explain your answer.

[2 marks]

6

Turn over for the next question



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9

A spinning platform makes 16 revolutions in 3.0 minutes.

What is the mean angular speed of the platform?

[1 mark]

A 0.090 rad s^{-1}

☐

B 0.28 rad s^{-1}

☐

C 0.56 rad s^{-1}

☐

D 33.5 rad s^{-1}

☐

A coin of mass 35 g is placed on a turntable turning horizontally at 120 revolutions per minute.

If the maximum force of friction between the coin and the turntable is 0.4 N, calculate the maximum radius for the coin to stay in circular motion.

[1 mark]

A 7.2 cm



B 7.2 cm



C 4.6 cm



D 4.6 m

